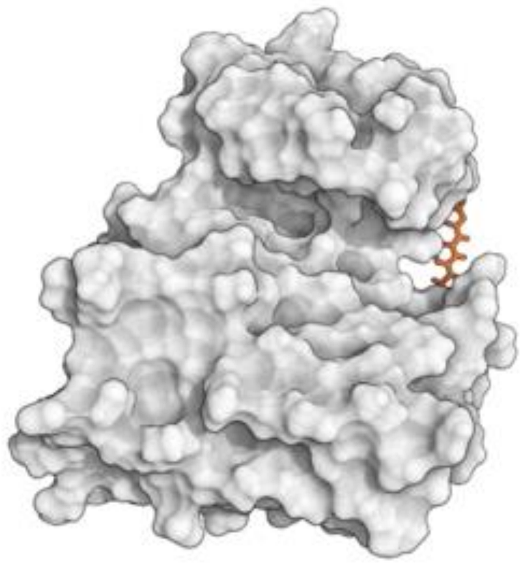
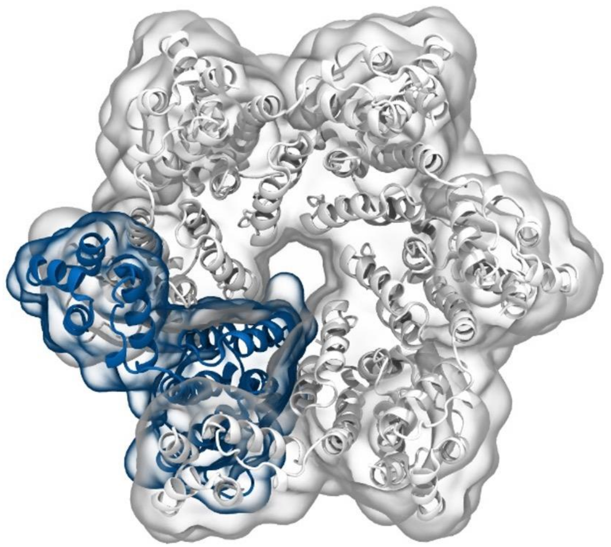
Simulation of Biomolecules



Setting up a protein simulation



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A typical workflow for molecular dynamics

Get protein structure

↓

Prep for simulation

↓

Simulation setup

↓

Run Simulation Protocol

↓

Simulation Analysis

Getting your protein structure

AlphaFold Protein Structure Database

RCSB PDB PROTEIN DATA BANK

Getting ligands/co-factors

ZINC20

RCSB PDB PROTEIN DATA BANK

A typical workflow for molecular dynamics

Get protein structure

↓

Prep for simulation

↓

Simulation setup

↓

Run Simulation Protocol

↓

Simulation Analysis

Prepping your protein and/or ligand(s)

SCHRÖDINGER Maestro

SeeSAR The drug design dashboard

[…]

A typical workflow for molecular dynamics

Get protein structure

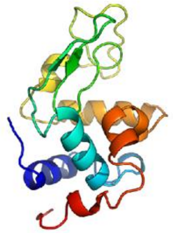
Prep for simulation

Simulation setup

Run Simulation Protocol

Simulation Analysis

Model of structure + **Model of energy of system**



$$U = \sum_{\text{All Bonds}} \frac{1}{2} K_b (b - b_0)^2$$

Your favourite simulation engine

Dynamics

A typical workflow for molecular dynamics

Get protein structure

Prep for simulation

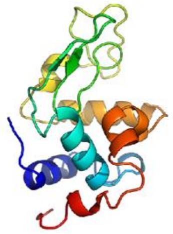
Simulation setup

Run Simulation Protocol

Simulation Analysis

All setup + protocol **decisions** are
driven by the question you want to answer

Model of structure



+

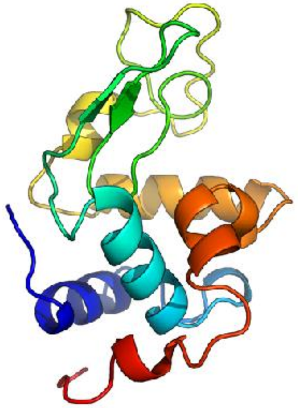
Model of energy of system

$$\underbrace{U = \sum \tfrac{1}{2} K_b (b-b_0)^2}_{\text{All Bonds}}$$

Your favourite simulation engine

Dynamics

Molecular dynamics require multiple steps for the setup of simulations



Disulphide bridges

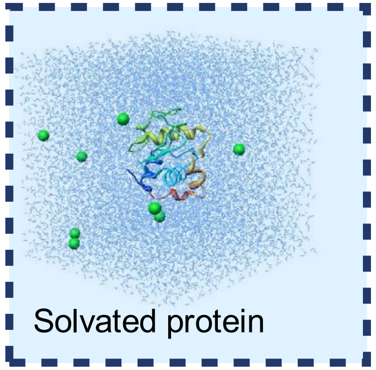
Protonation

Crystal Waters

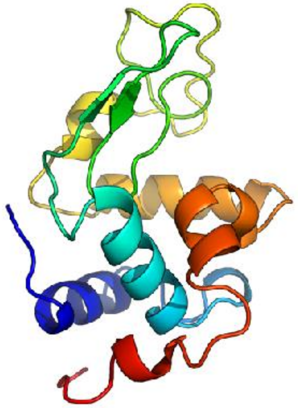
Cofactors

Alternative side chains

Solvate in water and add ions (*typically, neutralize the system*)



Molecular dynamics require multiple steps for the setup of simulations



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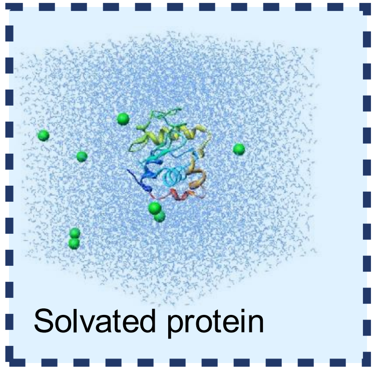
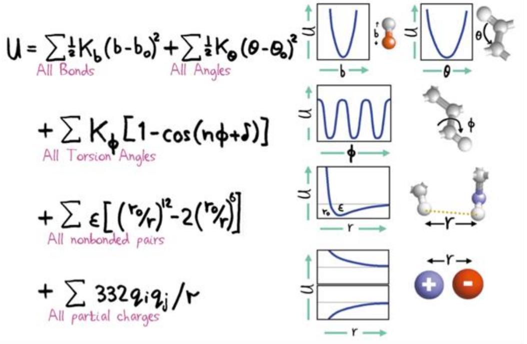


Apply forcefield description

$$\vec{F}_i = -\frac{dU}{d\vec{x}_i}$$

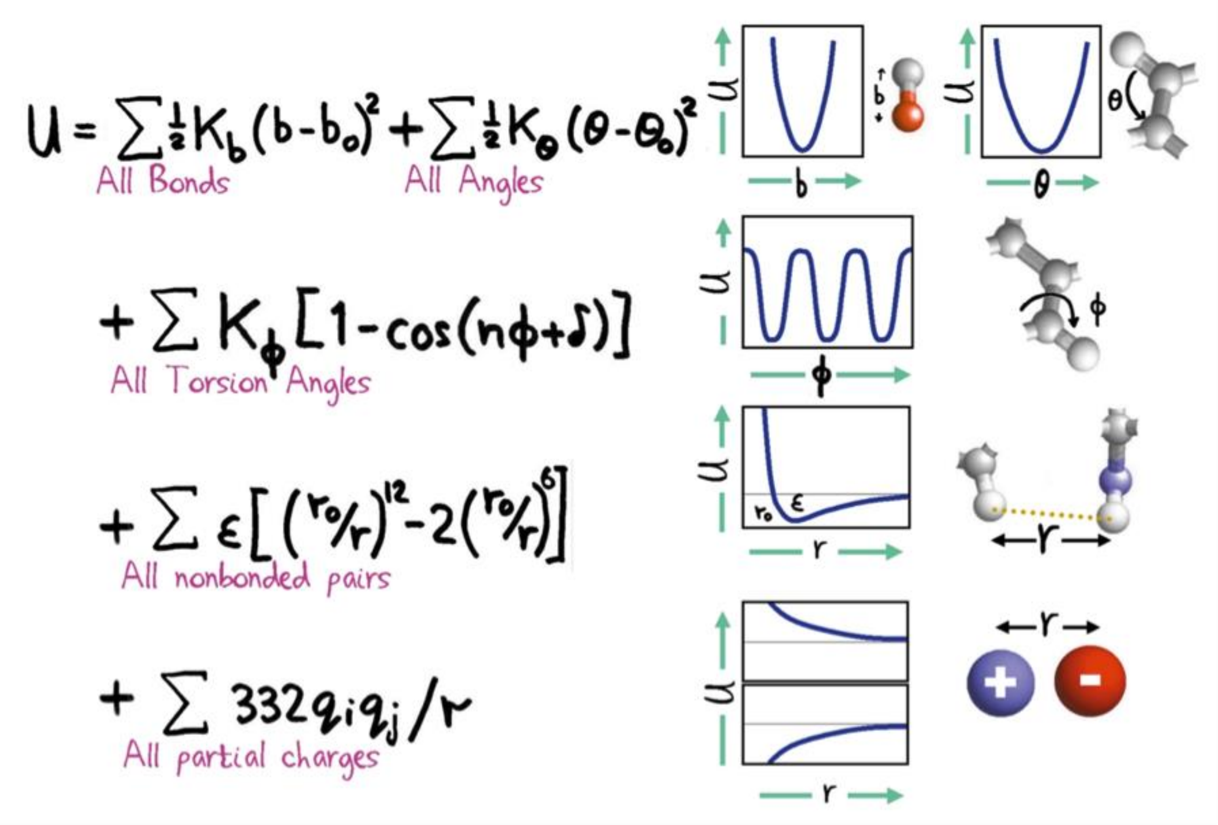


Levitt, ***Nature Struct. Biol.***, 2001

Molecular mechanics force fields balance accuracy and speed

- Often interested in relatively long-timescale motions
- Requires fast energy function

$$\vec{F}_i = -\frac{dU}{d\vec{x}_i}$$

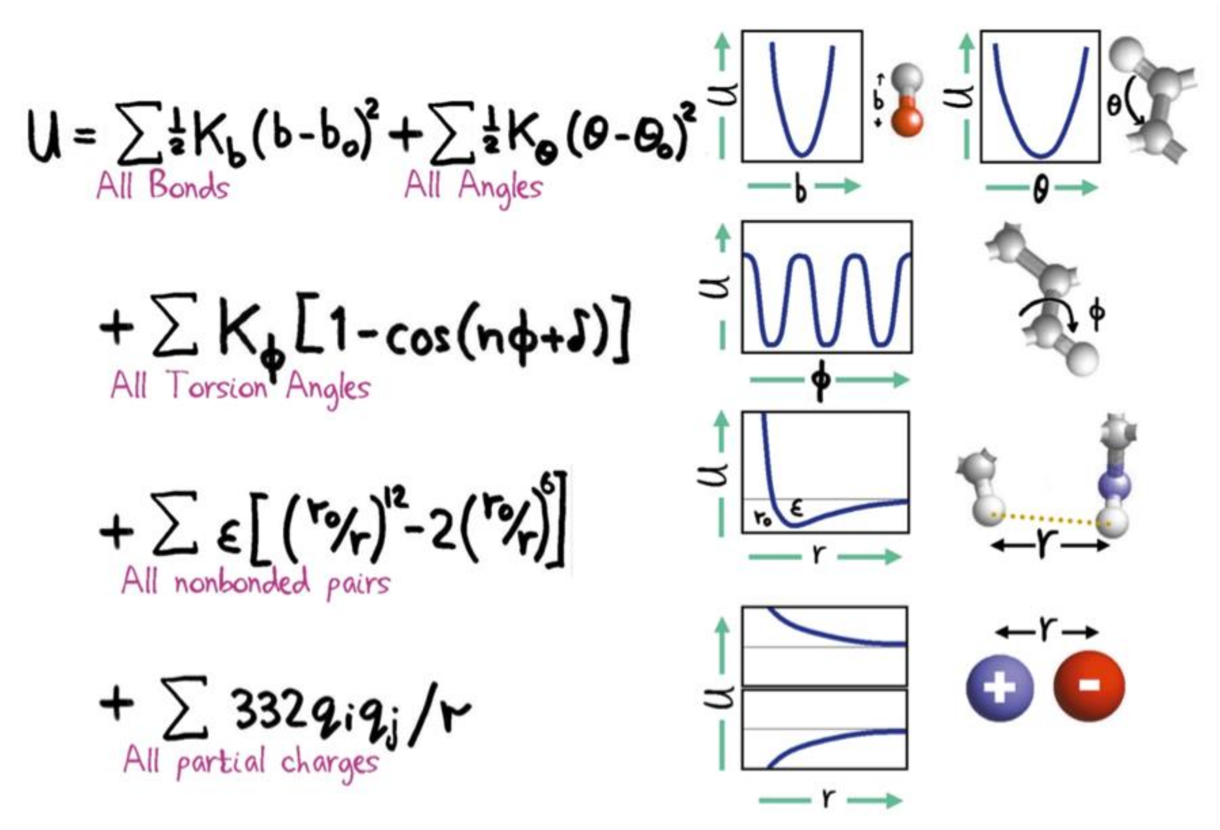


Levitt, ***Nature Struct. Biol.***, 2001

Molecular mechanics force fields balance accuracy and speed

- The functional form + **parameters** decide accuracy

$$\vec{F}_i = -\frac{dU}{d\vec{x}_i}$$

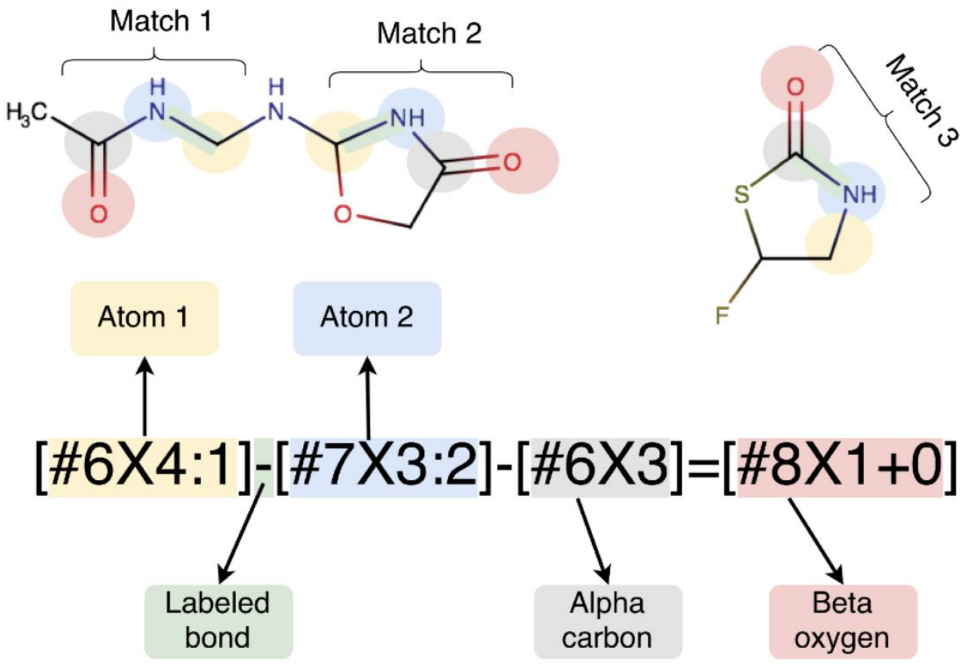


Levitt, ***Nature Struct. Biol.***, 2001

Molecular mechanics force fields balance accuracy and speed

Parameters are assigned using discrete types

SMIRKS Native Open Force Field Specification



Mobley et al., JCTC, 2018, **14**, 6076

Molecular mechanics force fields balance accuracy and speed

Parameters are assigned using discrete types

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<Bonds version="0.4" potential="harmonic" fractional_bondorder_method="AM1-Wiberg" fractional_bondorder_interpolation="linear">
    <Bond smirks="[#6X4:1]-[#6X4:2]" id="b1" length="1.525970013793 * angstrom ** 1" k="457.9258198725 * kilocalorie_per_mole ** 1
```

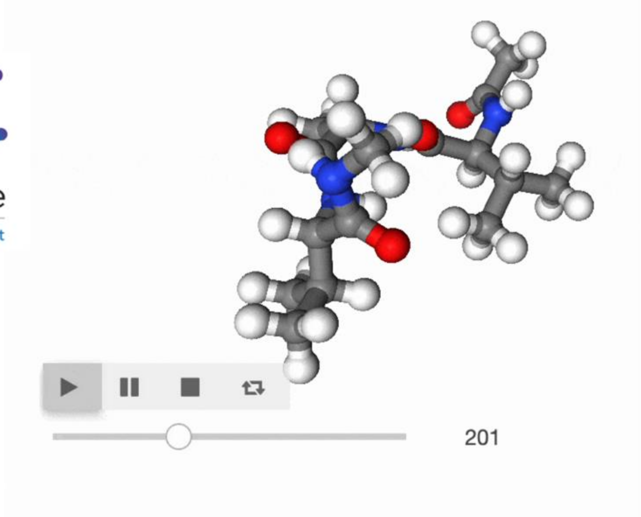
$$U = \sum_{\text{All Bonds}} \frac{1}{2} K_b (b - b_o)^2$$

Molecular mechanics force fields balance accuracy and speed

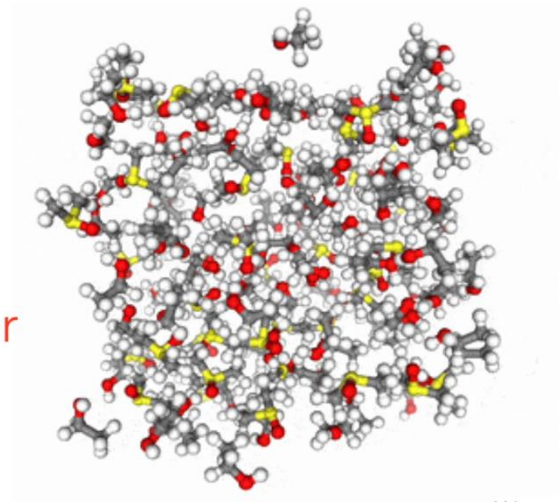
Parameters are fit to QM + experimental data

Bonded parameters trained on curated QM energies and forces of 1000's of drug-like molecules and fragments

QC Archive
A MolSSI Project



openff
evaluator



Non-bonded training set includes 1000's of mixture enthalpy and density measurements from NIST ThermoML

Wang L, Behara PK, et al. *Journal of Physical Chemistry B*, 2024, **128**, 7043-7067
Boothroyd S et al., *Journal of Chemical Theory and Computation*, 2023, **19**, 3251-3275

Molecular mechanics force fields balance accuracy and speed

Historically development has been messy…

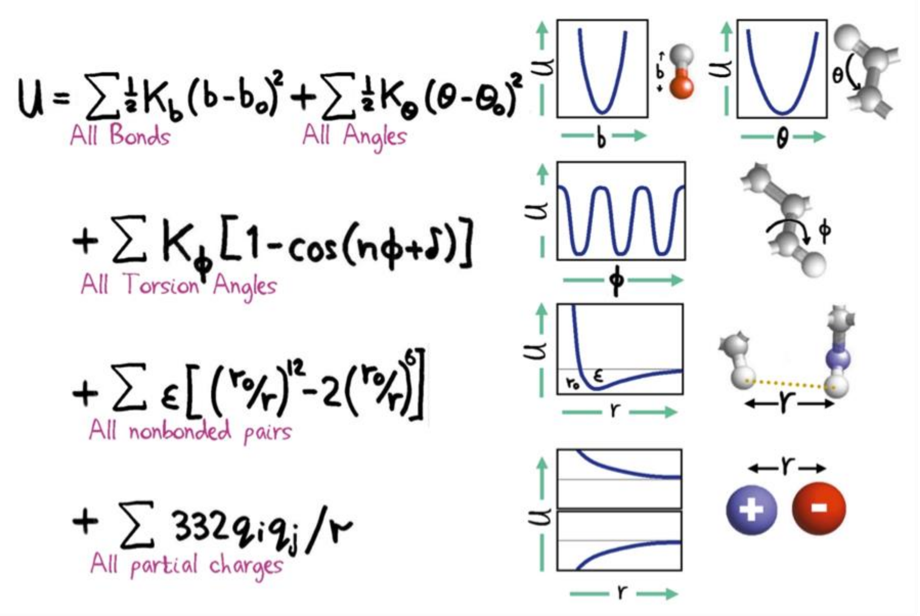
Listening to Mackerell talk about fitting Drude FFs he mentioned that they do normalization (during fitting) according to the phase of the moon... 😆 😂

“Force fields are like sausages; if you like them, it’s better not to know how they’re made”

Many force fields are available

Biomolecular force fields

$$\vec{F}_i = -\frac{dU}{d\vec{x}_i}$$



- **Amber** (incl. Glycam params for glycans)
- **CHARMM** (incl. POPC, POPE, DPPC lipids)
- **OPLS**
- **GROMOS**
- **…**

Small molecule force fields

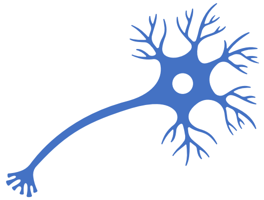
CGenFF

open forcefield

$$E_{pair} = \sum_{bonds} K_r (r - r_{eq})^2 + \sum_{angles} K_\theta (\theta - \theta_{eq})^2 + \sum_{dihedrals} \frac{V_n}{2}[1 + \cos(n\phi - \gamma)] + \sum_{i<j}\left[\frac{A_{ij}}{R_{ij}^{12}} - \frac{B_{ij}}{R_{ij}^{6}} + \frac{q_i q_j}{\varepsilon R_{ij}}\right]$$

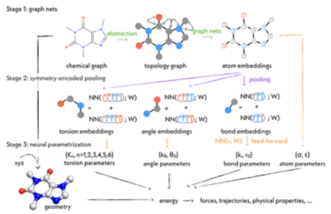
GAFF

Machine learned force fields



Ani-2x

Espaloma



SchNet

PhysNet

BandNN

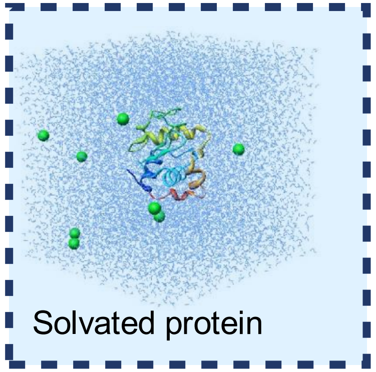
MACE-OFF

There is no "best force field"!

Many force fields are available … but not all are compatible

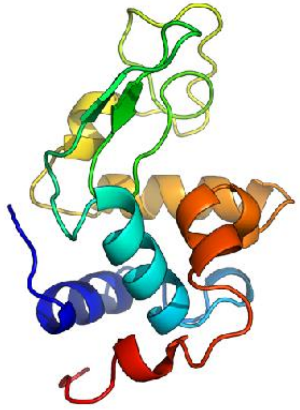
Be careful to use compatible water models and charge models etc…

AMBER ff14SB <-> TIP3P water
AMBER ff19SB <-> OPC water



There is no "best force field"!

Molecular dynamics require multiple steps for the setup of simulations



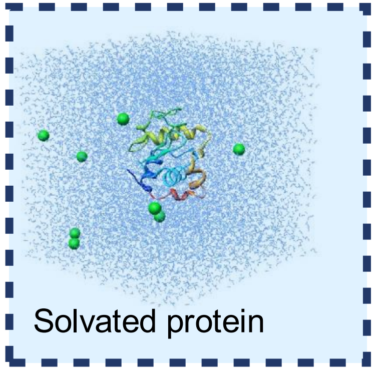
Disulphide bridges

Protonation

Crystal Waters Cofactors

Alternative side chains

Solvate in water and add ions (*typically, neutralize the system*)



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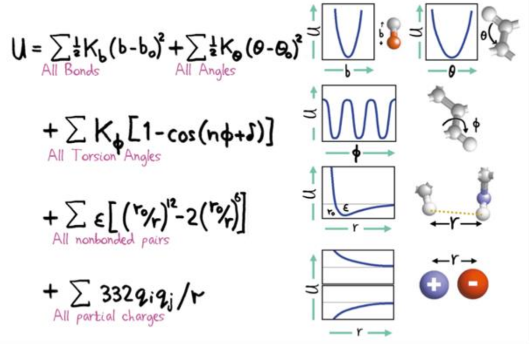
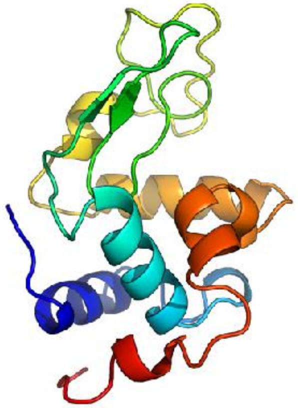


Levitt, ***Nature Struct. Biol.***, 2001

Molecular dynamics require multiple steps for the setup of simulations



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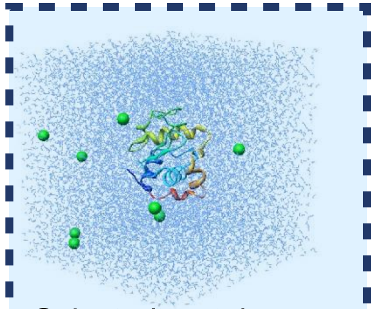
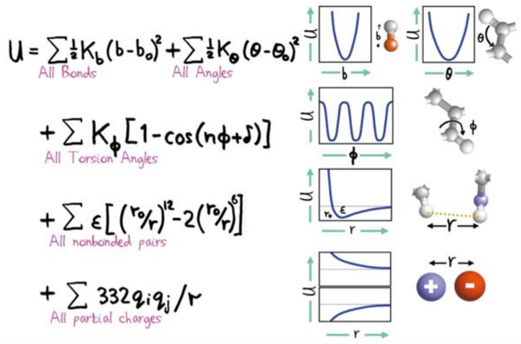
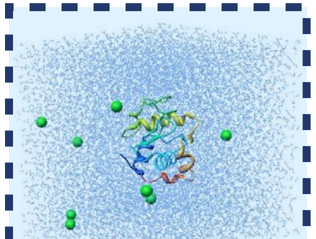


Solvated protein

Apply forcefield description

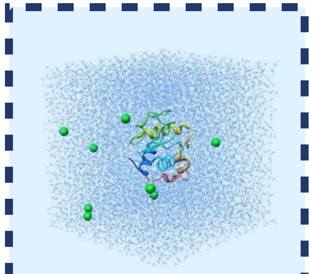
$$\vec{F}_i = -\frac{dU}{d\vec{x}_i}$$





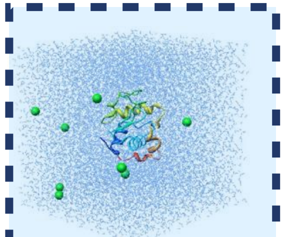
Solvated protein + forcefield

Minimize the positions

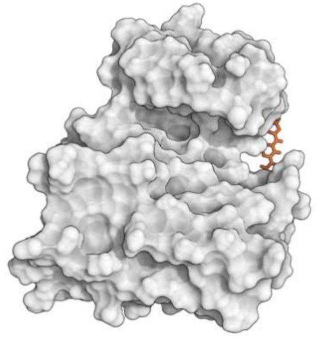


minimized protein

NVT/NPT equilibration



Equilibrated system



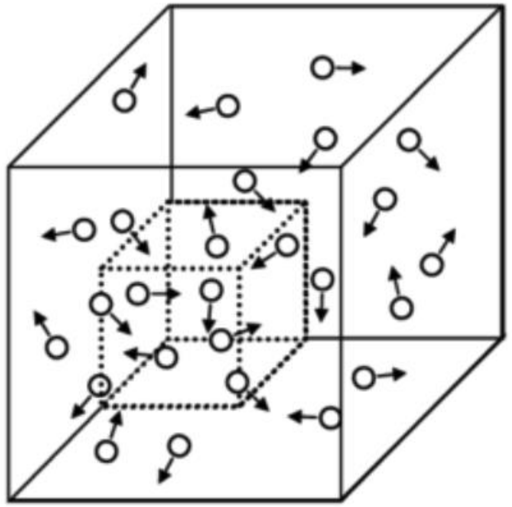
Production simulation

http://www.bevanlab.biochem.vt.edu/Pages/Personal/justin/gmx-tutorials/lysozyme/01_pdb2gmx.html

Levitt, ***Nature Struct. Biol.***, 2001

Choosing your thermodynamic ensemble

Simulations replicate a specific *thermodynamic ensemble* (typically NVT or NPT), or even grand canonical (μVT)



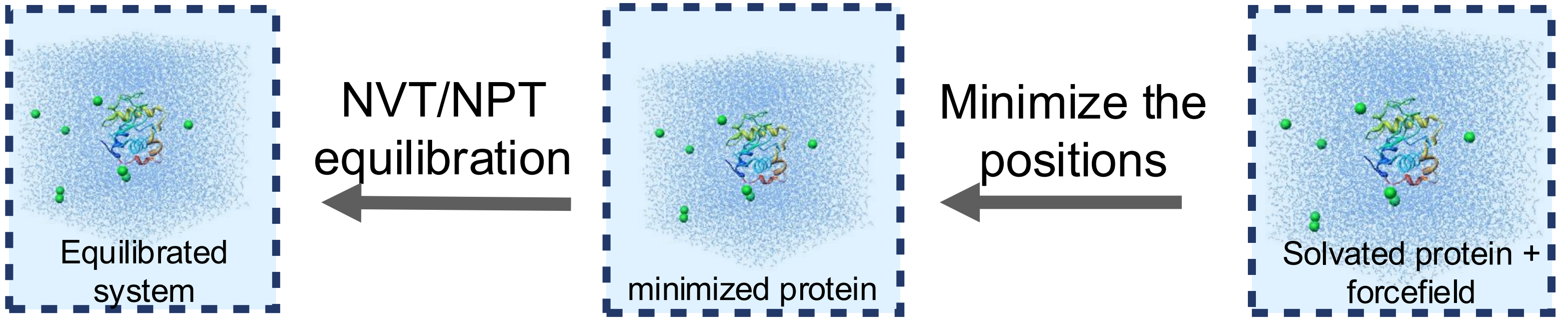
μVT

You will have different options to include *thermostats* (scaling atom velocities) and *barostats* (scaling positions) in your calculations:

- Nose-Hoover
- Berendsen
- Parrinello-Rahman
- Langevin piston
- …

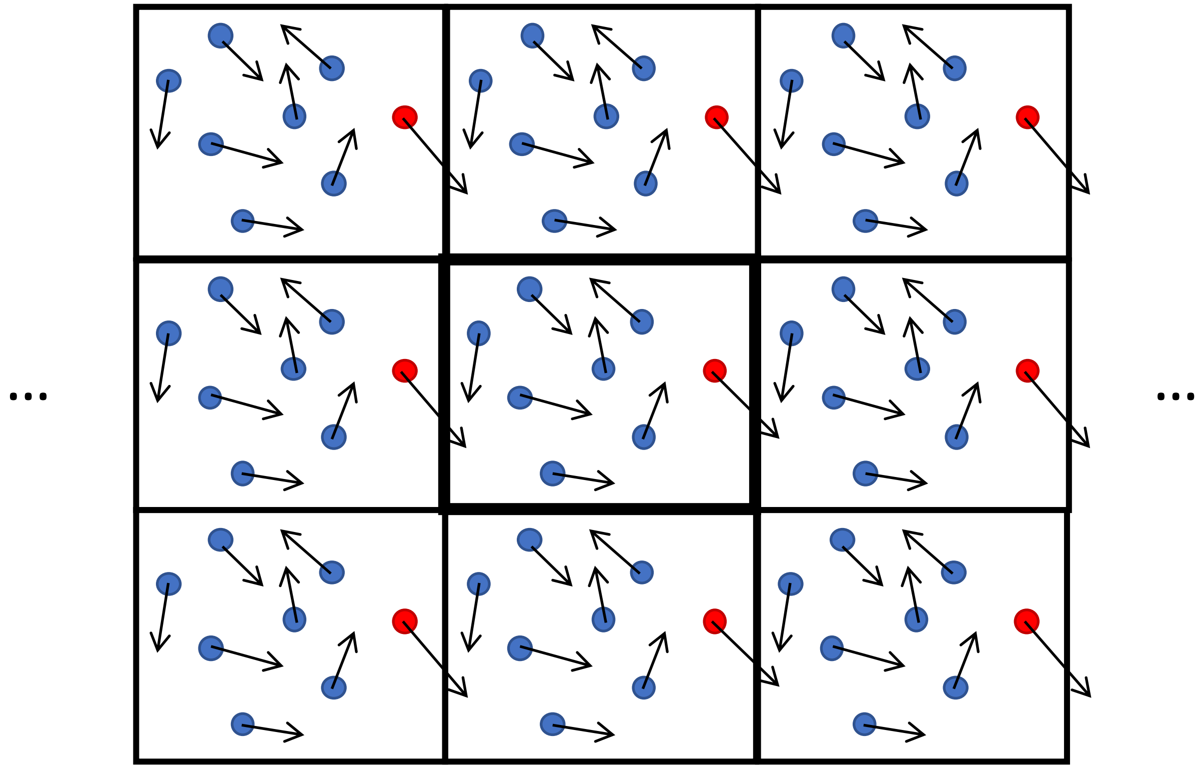
Best Practices for Foundations in Molecular Simulations [Article v1.0]

Efrem Braun[1], Justin Gilmer[2], Heather B. Mayes[3], David L. Mobley[4], Jacob I. Monroe[5], Samarjeet Prasad[6], Daniel M. Zuckerman[7]

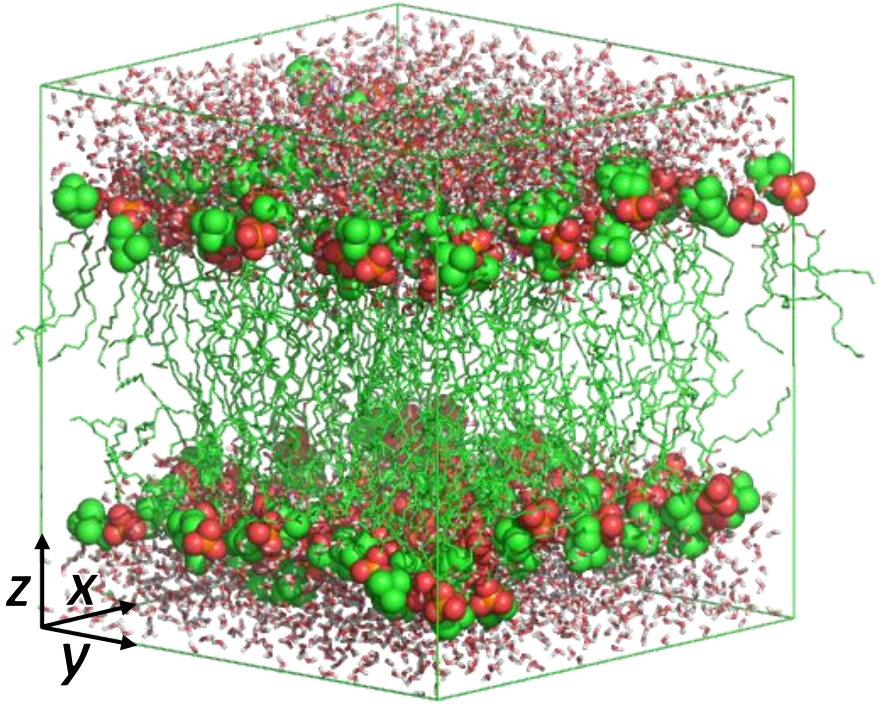


Periodic boundary conditions (PBC) and pressure coupling

Useful to reduce finite-size effect and simulate bulk



Typically, PBC applied in x, y and z direction



For membrane systems, use semi-isotropic pressure coupling (x, $y \neq z$, lipids compressibility is direction-dependent)

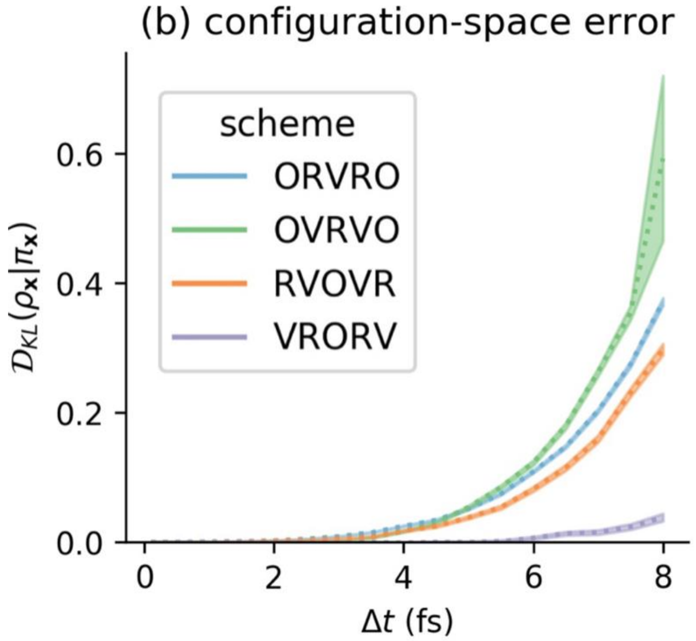
Sampling timescales for protein systems

The fastest vibration determines the smallest timestep.

Timestep size is imposed by the fastest phenomenon we want to observe.
In atomistic simulations:

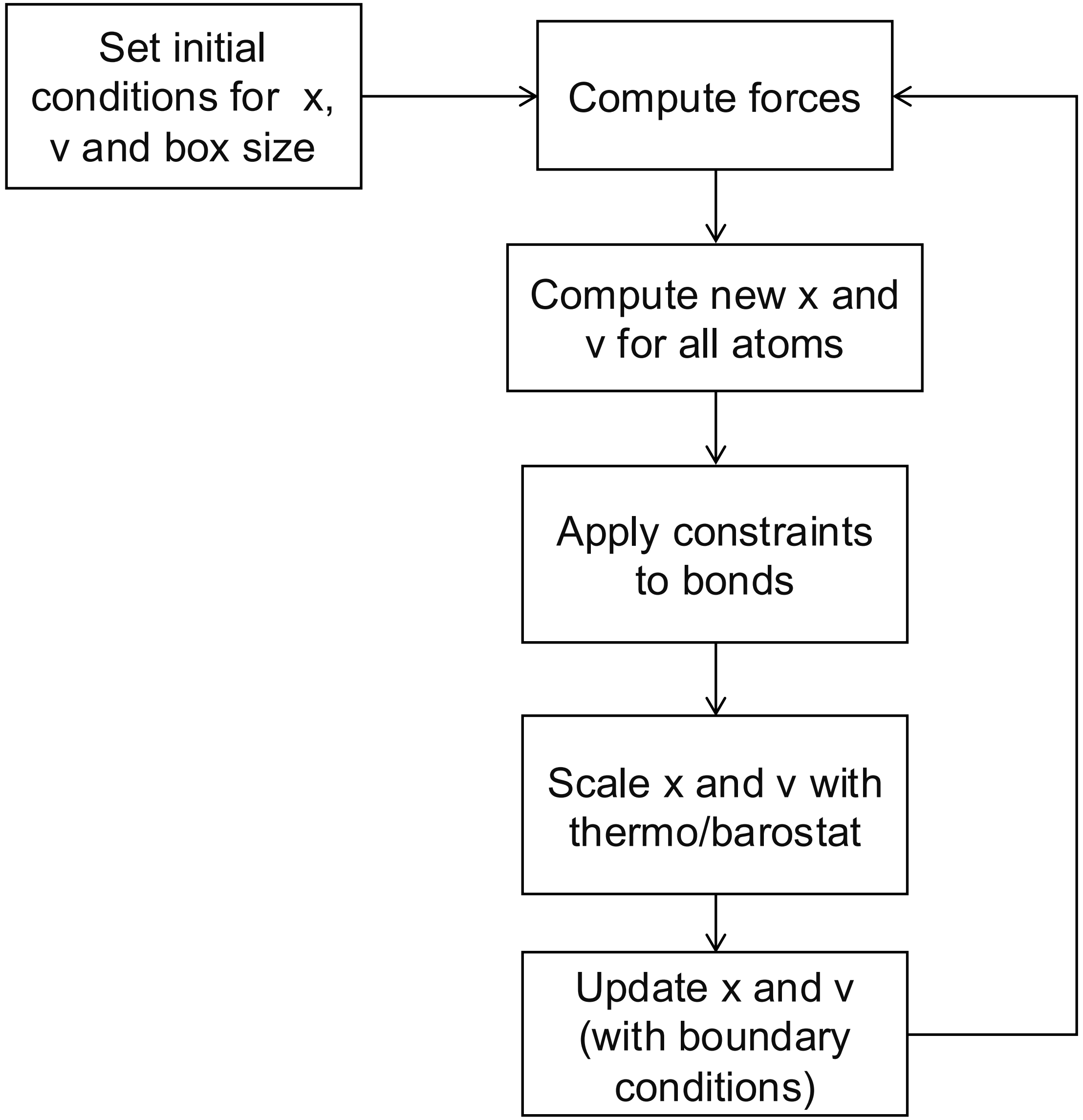
- Covalent bond hydrogen-heavy atom (10^{14} Hz): 0.5 fs
- Covalent bond heavy atom-heavy atom: 1 fs
- Angles fluctuations: 2 fs

Restraining covalent bond distances allows to use 1-2 fs timesteps (restraining methods: SHAKE, RATTLE, LINCS,…)
Hydrogen Mass repartitioning: 4 fs
Other integrators (e.g., Langevin): 4 fs - 6 fs.



https://doi.org/10.1101/266619

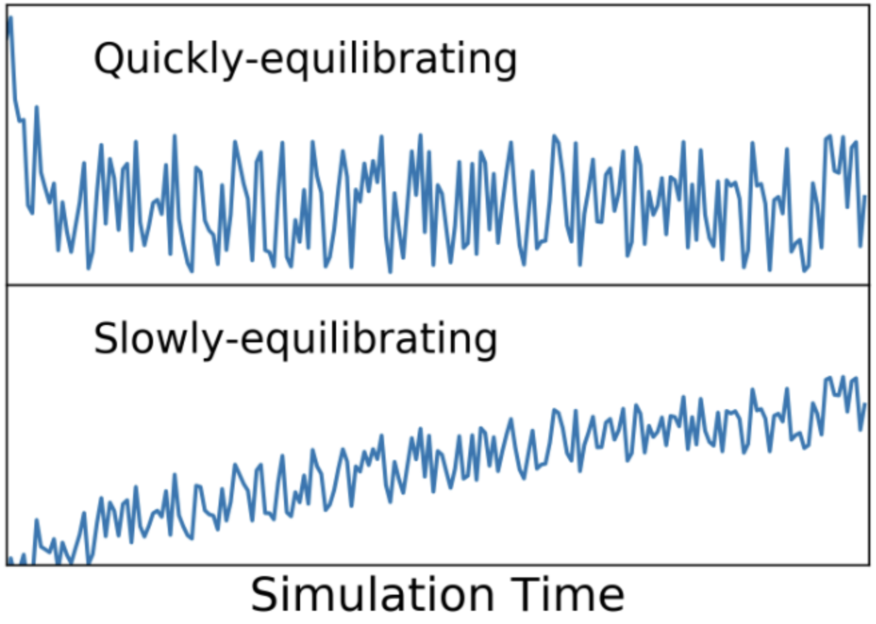
A Molecular Dynamics timestep



“equilibration” and “convergence”: what do they mean?

“Equilibration”:

- Relaxation from starting configuration which has very low equilibrium probability in your ensemble
- “Unequilibrated” data usually discarded
- Related to when you start collecting data



https://doi.org/10.33011/livecoms.1.1.5957

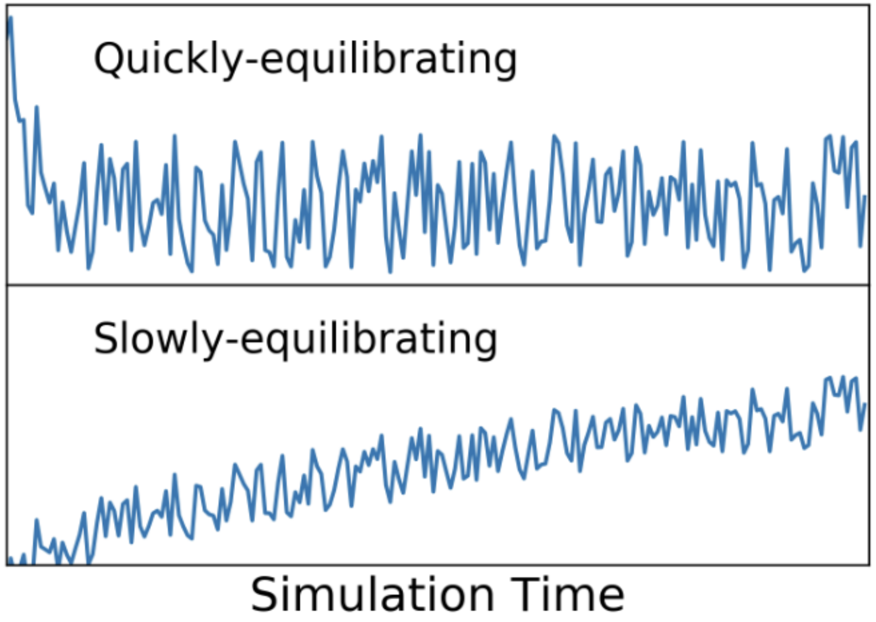
Best Practices for Quantification of Uncertainty and Sampling Quality in Molecular Simulations [Article v1.0]

https://doi.org/10.33011/livecoms.1.1.5067

“equilibration” and “convergence”: what do they mean?

“Convergence”:

- Extent to which an estimator approaches some “true” value (expectation of observable) with increasing amounts of data
- Often approximated by agreement of replicates
- Related to when you stop collecting your data



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Best Practices for Quantification of Uncertainty and Sampling Quality in Molecular Simulations [Article v1.0]

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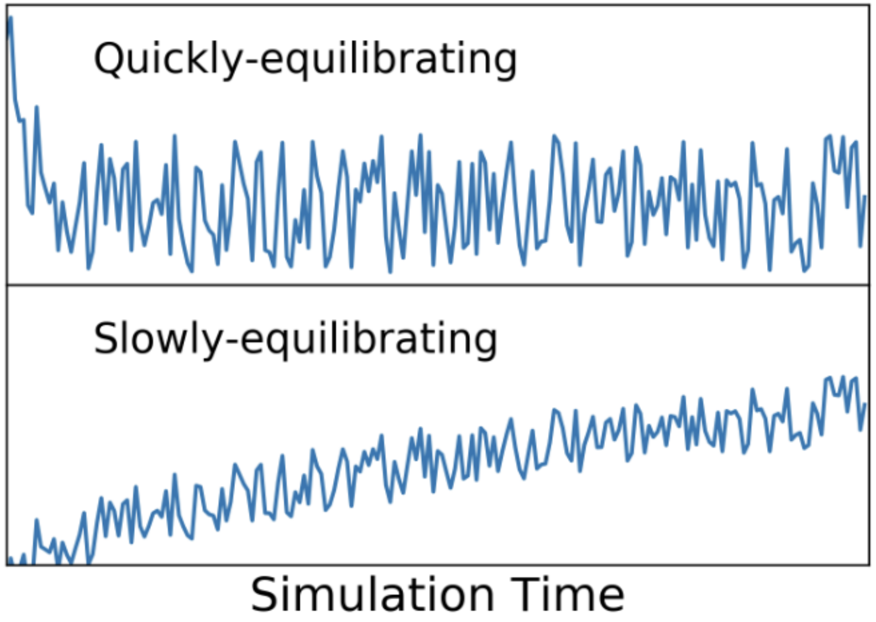
"equilibration" and "convergence": what do they mean?

- Different quantities will "equilibrate" and "converge" at different rates
- Sometimes obvious we have not "converged/ equilibrated", but almost never that we have
- Running independent replicates is best practice for uncertainty quantification



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Best Practices for Quantification of Uncertainty and Sampling Quality in Molecular Simulations [Article v1.0]

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Example equilibration protocols

YOU WANT: constant volume, pressure, and temperature, healthy Ramachandran plot, no exotic chemistry, bulk water (if used)

Suggested equilibration workflow				Production Ensemble
NVT (short simulation to relax to temperature of interest) →			NVE (short equilibration) →	NVE
NVT (short simulation to relax to temperature of interest) →				NVT (at known, fixed density)
NVT (short simulation to relax to temperature of interest) →	NPT (short simulation to relax to density of interest) →	NPT (to calculate average box size) →	NVT (short equilibration) →	NVT (for density defined by pressure or unknown system density distribution, like a homegeneous system)
NVT (short simulation to relax to temperature of interest) →	NPT (short simulation to relax to density of interest) →			NPT

*Living J. Comp. Mol. Sci. 2019, **1**, 5957*

An *example* simulation protocol

YOU WANT: constant volume, pressure, and temperature, healthy Ramachandran plot, no exotic chemistry, bulk water (if used)

Equilibration:

1. Minimize energy, 1000 steepest descent
2. Heat system from 0 to 300 K in 500 ps, NPT, Berendsen barostat 1 atm. α-carbon restrained with 10 kcal/mol harmonic potential. 2 fs timestep, LINCS all bonds
3. 1 ns NVT equilibration with Langevin dynamics, no atom constrained.

Production:

4. 1 µs NPT, Nose-Hoover barostat, PME for electrostatics

DETERMINE HANDLING OF CUTOFFS

- ☐ As a general rule, electrostatics are long-range enough that either the cutoff needs to be larger than the system size (for finite systems) or periodicity is needed along with full treatment of long-range electrostatics (Section 3.4)
- ☐ Nonpolar interactions can often be safely treated with cutoffs of 1-1.5 nm as long as the system size is at least twice that, but long-range dispersion corrections may be needed (Section 4.1)

CHOOSE APPROPRIATE SETTINGS FOR THE DESIRED ENSEMBLE

- ☐ Pick a thermostat that gives the correct distribution of temperatures, not just the correct average temperature; if you have a small system or a system with weakly interacting component choose one which works well even in the small-system limit.
- ☐ Pick a barostat that gives the correct distribution of pressures
- ☐ Consider the known shortcomings and limitations of certain integrators and thermostats/barostats and whether your choices will impact the properties you are calculating

CHOOSE AN APPROPRIATE TIMESTEP FOR STABILITY AND AVOIDING ENERGY DRIFT

- ☐ Determine the highest-frequency motion in the system (typically bond vibrations unless bond lengths are constrained)
- ☐ As a first guess, set the timestep to approximately one tenth of the highest-frequency motion's characteristic period
- ☐ Test this choice by running a simulation in the microcanonical ensemble, and ensure that energy is conserved

*Living J. Comp. Mol. Sci. 2019, **1**, 5957*

A typical workflow for molecular dynamics

Get protein structure

↓

Prep for simulation

↓

Simulation setup

↓

Run Simulation Protocol

↓

Simulation Analysis

Setup and run on your favourite MD engine

Gromacs

OpenMM

Amber

NAMD

BioSimSpace

SOMD

Charmm

DLPoly

LAMMPS

mostly C++
command line

Python API

Which MD engine should I use?

Consider:

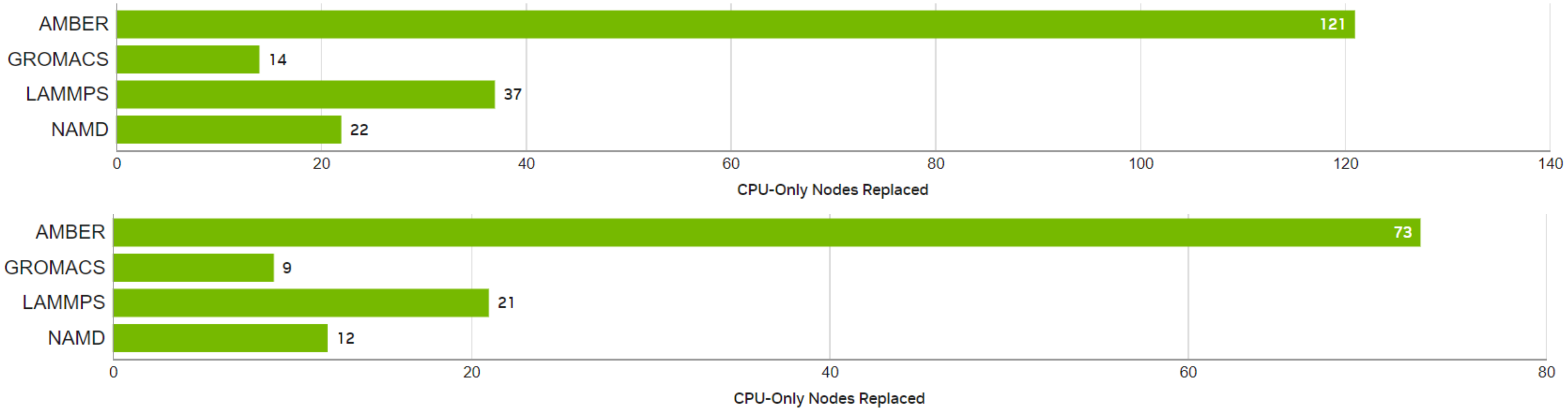
- Support for force field of choice
- Enables running desired simulation protocol
- Performance for available hardware → *depends on number of atoms, hardware, simulation protocol, MD engine*
- Ease of use

Graphical Processing Units (GPUs) are especially effective for MD

From: developer.nvidia.com/hpc-application-performance

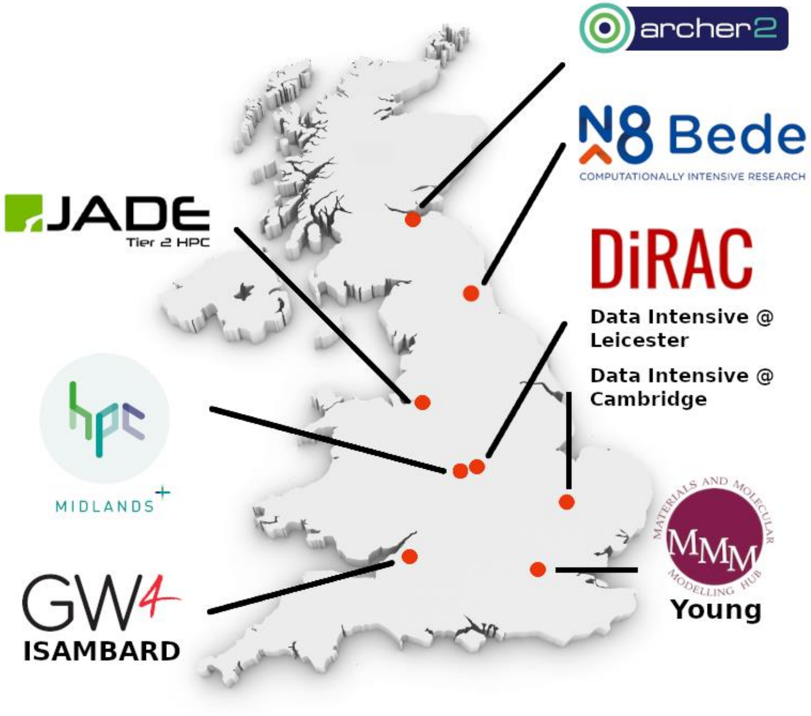
NVIDIA H200

NVIDIA A100

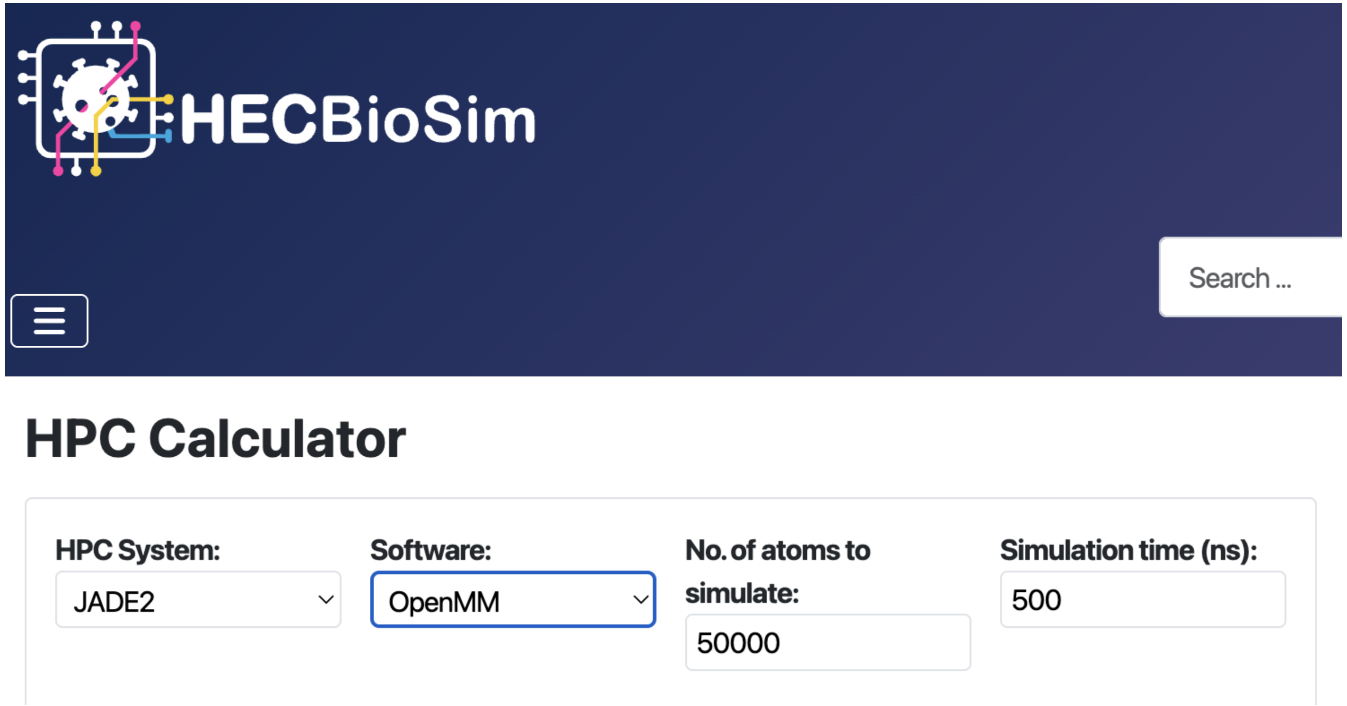


CPU Server: Dual Xeon Platinum 8480+ @2GHz | GPU Server: Dual EPYC 7742@2.25GHz with 4x NVIDIA A100 SXM 80GB | AMBER Benchmark: DC-Cellulose_NVE, CUDA Version: 12.2 | GROMACS Benchmark: ADH Dodec, CUDA Version: 12.4 | LAMMPS Benchmark: Tersoff, CUDA Version: 12.2 | NAMD Benchmark: apoa1_nve_cuda, CUDA Version: 12.2

Calculating runtimes: example on UK Tier 2 systems

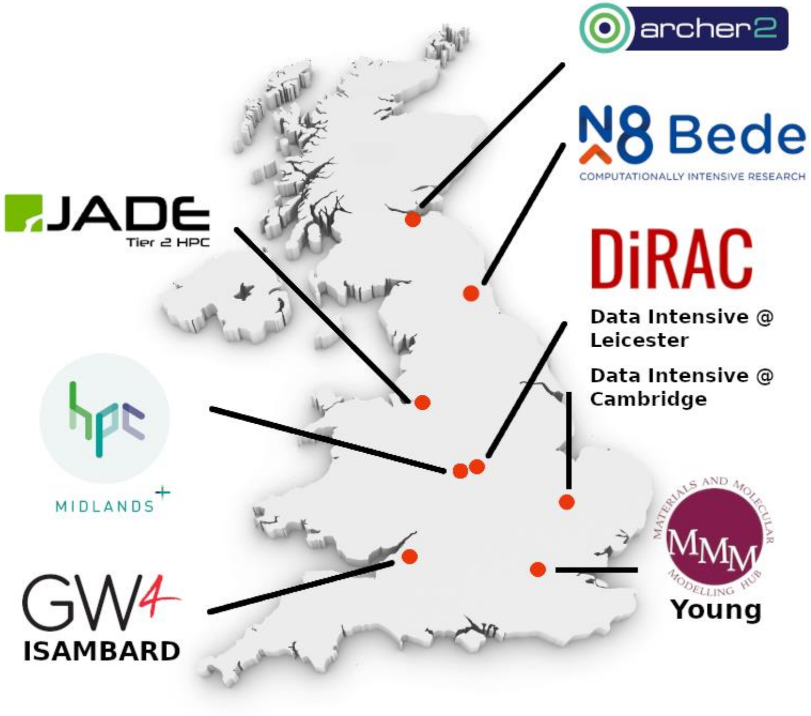


[as of 2024]



https://www.hecbiosim.ac.uk/access-hpc/hpc-calculator

Calculating runtimes: example on UK Tier 2 systems



[as of 2024]

HECBioSim

Search ...

HPC Calculator

HPC System: JADE2

Software: OpenMM

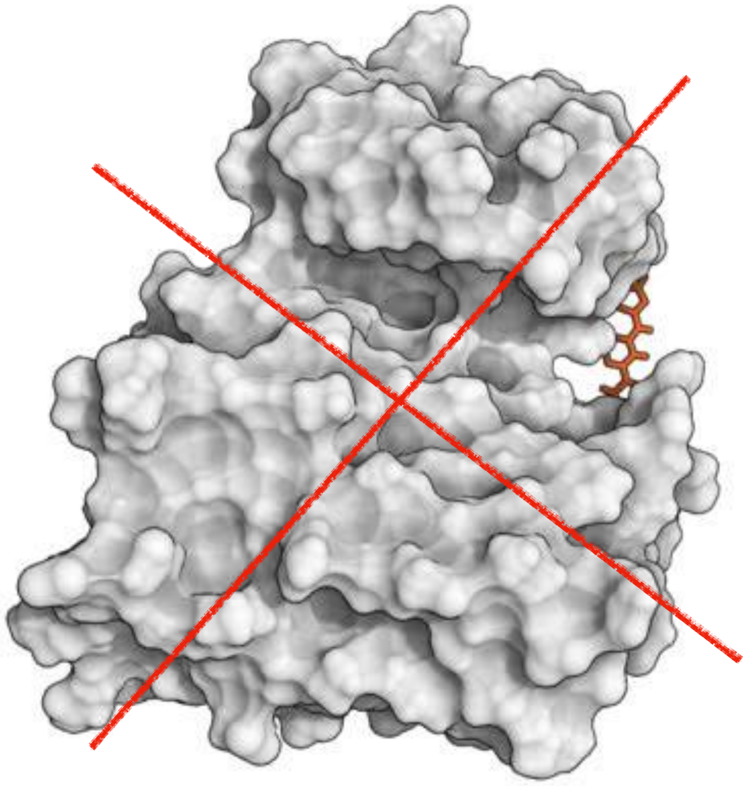
No. of atoms to simulate: 50000

Simulation time (ns): 500

179 ns/day
47.5GB of storage
67 node hours
1 node recommended for maximum efficiency
33.2 kWh of power
...which could power an average UK home for 4.49 days

https://www.hecbiosim.ac.uk/access-hpc/hpc-calculator

Disclaimer!



Running biomolecular MD can take days on specialised hardware.

Today we will *not* run any of them, and instead will focus on fundamental principles using small molecules.

Workshop 4: OpenFF Tooklit + Interchange

OPEN SOFTWARE

Automated infrastructure enables rapid experimentation with minimum human intervention

OPEN DATA

Access to large, high quality experimental and quantum chemical data facilities easy curation of balanced train / test sets

OPEN SCIENCE

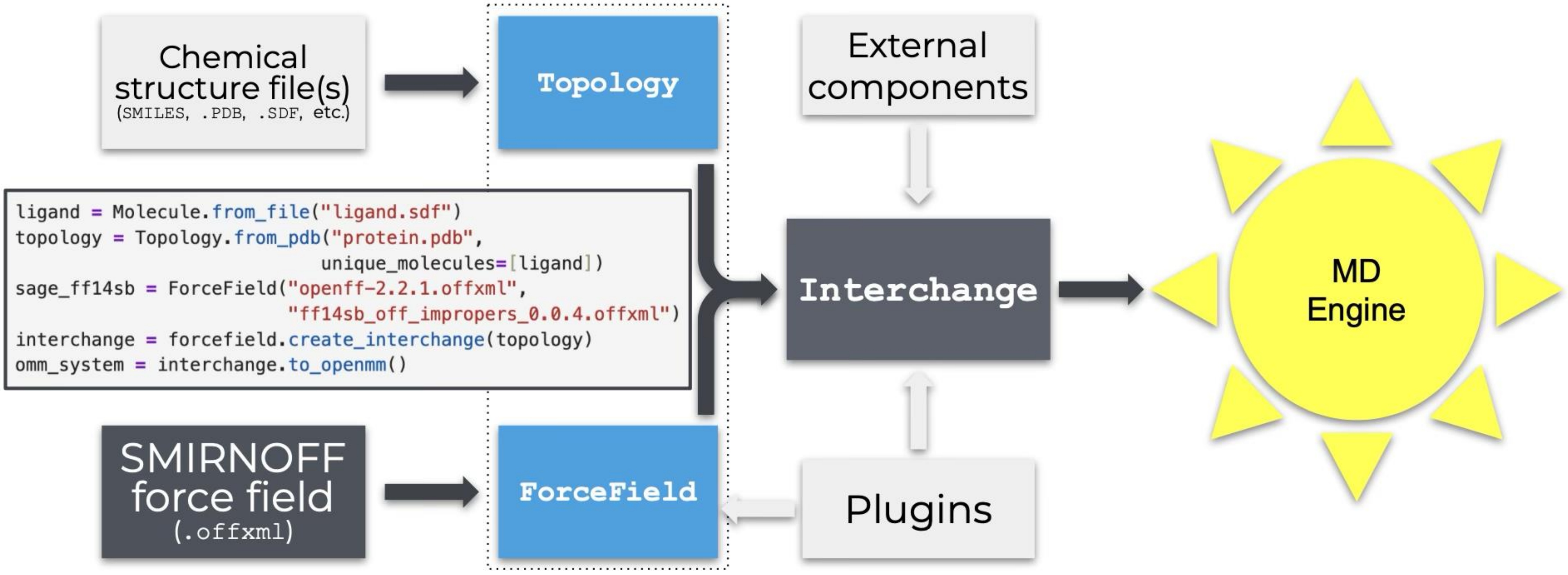
Exploring new force field science:

hypothesis - build software - train - test - iterate

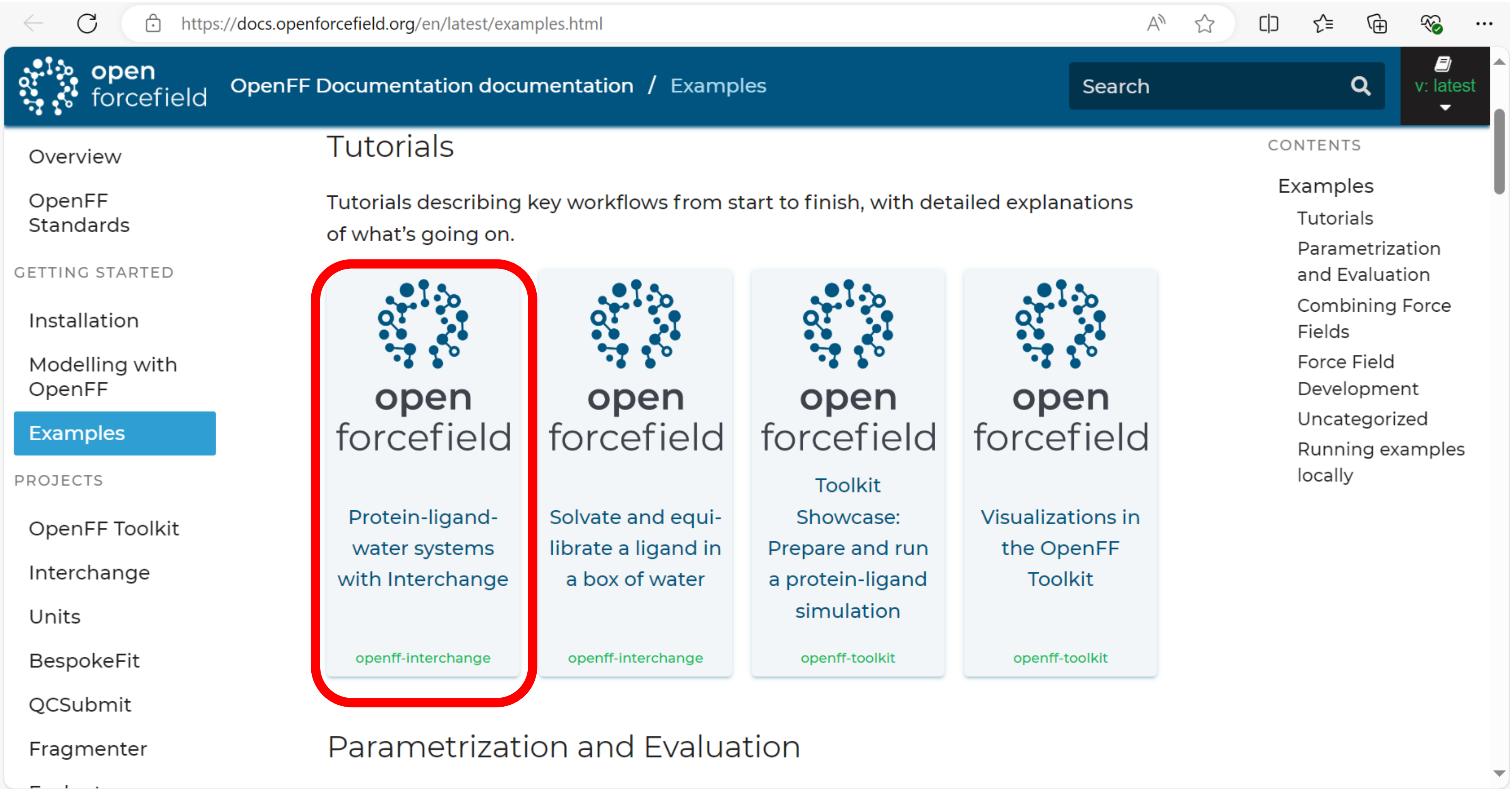
is now almost routine

https://openforcefield.org/

Workshop 4: OpenFF Tooklit + Interchange



Workshop 4: OpenFF Tooklit + Interchange



https://docs.openforcefield.org/en/latest/examples.html

Useful resources to learn running simulations

Best Practices for Foundations in Molecular Simulations [Article v1.0]

Efrem Braun[1], Justin Gilmer[2], Heather B. Mayes[3], David L. Mobley[4], Jacob I. Monroe[5], Samarjeet Prasad[6], Daniel M. Zuckerman[7]

A suite of tutorials for the BioSimSpace framework for interoperable biomolecular simulation [Article v1.0]

Lester O. Hedges[1,2*], Sofia Bariami[3†], Matthew Burman[2], Finlay Clark[3], Benjamin P. Cossins[4], Adele Hardie[3], Anna M. Herz[3], Dominykas Lukauskis[5], Antonia S.J.S. Mey[3], Julien Michel[2,3*], Jenke Scheen[3‡], Miroslav Suruzhon[4], Christopher J. Woods[1], Zhiyi Wu[4]

From Proteins to Perturbed Hamiltonians: A Suite of Tutorials for the GROMACS-2018 Molecular Simulation Package [Article v1.0]

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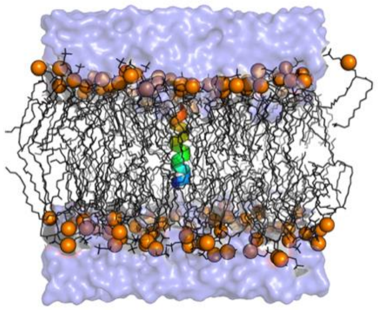

DOI: https://doi.org/10.33011/livecoms.1.1.5068

Keywords: tutorials, gromacs, molecular dynamics simulation, computational chemistry

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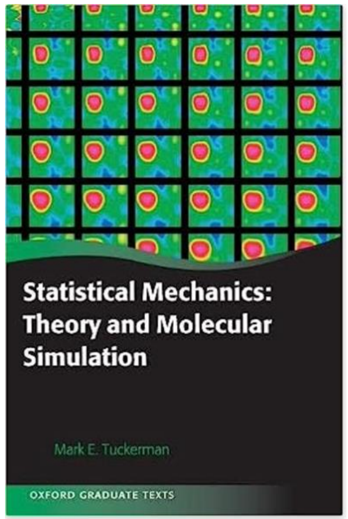

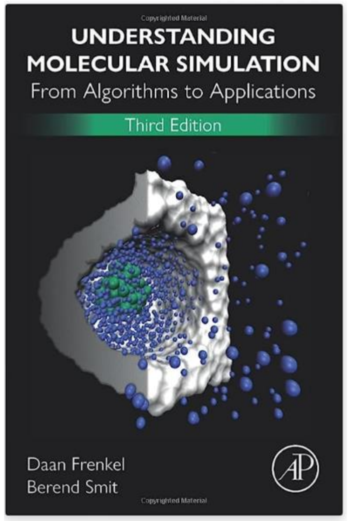

GROMACS: tutorials.gromacs.org

Amber: ambermd.org/tutorials

OpenMM: https://openmm.github.io/openmm-cookbook/dev/index.html